STSCI 3080 : Lecture 1 (Chapter 1 in DeGroot and Schervish)

- Probability (event) = How "likely" the event is to occur?

Definition 1 (Experiment and event)

- Experiment = Any process, real or hypothetical, in which the possible outcomes can be identified ahead of time.
- Event = A well defined set of possible outcomes of the experiment.

Example 1

Experiment = Toss a coin ten times.

Possible event of interest = I see at least 4 heads.

Example 2

Experiment = Observe the lifetime of a TV.

Possible event of interest = TV doesn't last longer than 5 years.

Sample spaces, events and set theory

Definition 2 The collection of all possible outcomes of an experiment is called the sample space.

- The sample space $\longleftrightarrow$ Set
  - $\uparrow \psi$ $\qquad$ $\uparrow \psi$
- Outcome $\longleftrightarrow$ Point / Element

- Event $\longleftrightarrow$ Subset (of the sample space).

Example

- Sample space associated with tossing two dice:

$$S = \{(i,j);\ i, j \in \{1, 2, 3, 4, 5, 6\}\}.$$

- Event = sum of dice equals 3:

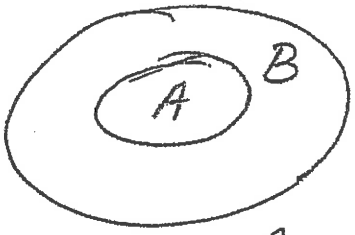
$$E = \{(1,2), (2,1)\} \subset S.$$

"Subset of"

- From now on Event = set.
- To study relationships between events or to build complex events from simpler ones we would therefore need to know how to perform operations with sets.

Review of basic set theory

Definition 1 $A \subset B$ iff $\forall x \in A \Rightarrow x \in B$.
(Every element in A is also in B).



- We also say B "contains" A or A "included" in B.

Definition 2 $A = B$ iff $A \subset B$ and $B \subset A$.

Definition 3 A set that contains no elements is called the empty set $\emptyset$.

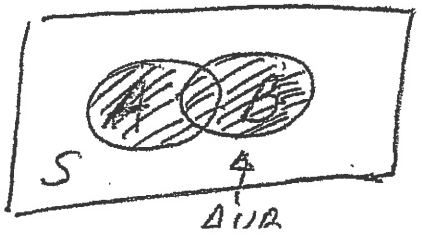
Definition 4.

- If a set has finitely many elements, then it is called a finite set.
- An infinite set is called countable if there is a one-to-one correspondence between its elements and the set of real numbers $\{1, 2, 3, \ldots, n, \ldots\}$.
- An infinite set is called uncountable if it is neither finite, nor countable.

Definition 5 Let $A \subset S$. Then A^c, the complement of A, is the set that contains all elements of S that are not in A.

Definition 6 If A and B are two sets, the union of A and B is the set that contains the elements of A alone, of B alone and those that belong to both A and B.

Notation: $A \cup B$



Definition 6' Let $A_1, \ldots, A_n$ be n sets.

Their union:

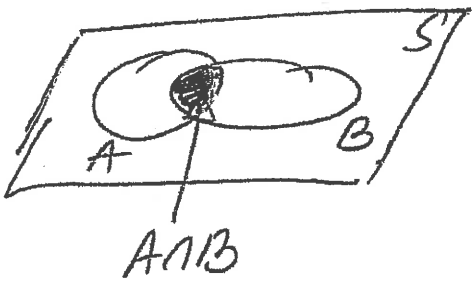
$$A_1 \cup A_2 \cup \ldots \cup A_n = \bigcup_{i=1}^{n} A_i \quad \text{is}$$

the set that contains elements that belong to at least one of these n sets.

Note If $A_1, \ldots, A_n, \ldots$ is a countable sequence of sets we use the notation below for their union:

$$\bigcup_{i=1}^{\infty} A_i.$$

Definition 7 (a) If A and B are two sets, their intersection $A \cap B$ is the set that contains elements common to <u>both</u> A and B.



(6) $\bigcap_{i=1}^{n} A_i = A_1 \cap \ldots \cap A_n$ contains elements that are common to all sets

Definition 8 Two sets A and B are called disjoint, or mutually exclusive, if they have no elements in common, that is $A \cap B = \emptyset$.

Properties of sets and of operations with sets

(1) $A \subset B$ and $B \subset C$, then $A \subset C$, for any A, B, C.

(2) $\emptyset \subset A$, for any A.

(3) $(A^c)^c = A$, for any $A \subset S$, for some S.

(4) For any sets A and B (assumed to be included in some generic set S) we have:

(a) $A \cup B = B \cup A$

(b) $A \cup A = A$

(c) $A \cup A^c = S$

(d) $A \cup \emptyset = A$

(e) $A \cup S = S$

(f) If $A \subset B$, then $A \cup B = B$.

(4') For any A, B, C we have:

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

(5) For any sets $A, B, C (\subset S)$ we have:

(a) $A \cap B = B \cap A$

(b) $A \cap A = A$

(c) $\boxed{A \cap A^c = \emptyset}$

(d) $A \cap \emptyset = \emptyset$

(e) $A \cap S = A$

(f) If $A \subset B$, then $A \cap B = A$

(g) $A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C)$.

VERY IMPORTANT PROPERTIES

(I) De Morgan's Laws:

$$[A \cup B]^c = A^c \cap B^c;$$

$$[A \cap B]^c = A^c \cup B^c.$$

(II) Partioning a set For any A and B:

$A = (A \cap B) \cup (A \cap B^c)$, where we notice hat $A \cap B$ and $A \cap B^c$ are disjoint sets.

?

(III) Distributive properties

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

for any three sets A, B and C.

Set-event dictionary

Events	Sets
The "certain" event = the sample space	S
An event "E".	$E \subset S$
The impossible event	$\emptyset$
If A occurs, so does B The occurence of A implies that of B).	$A \subset B$
A does <u>not</u> occur	A^c
Either A <u>or</u> B <u>or</u> both occur	$A \cup B$.

Events	Sets
At least one event in a collection of events occur.	$A_1 \cup A_2 \cup \ldots \cup A_n \ldots$
Both A and B occur.	$A \cap B$
Every event in a collection occur.	$A_1 \cap A_2 \cap \ldots \cap A_n \cap \ldots$
A and B cannot both occur (Cannot occur together).	$A \cap B = \emptyset$

Examples

- Experiment = Flip two coins
- Sample space = $S = \{(H,T); (T,H); (T,T); (H,H)\}$
- Possible events of interest:

$A = \{(T,T)\} \leftrightarrow$ Exactly 2 tails appear.

$B = \{(H,H)\} \leftrightarrow$ Exactly 2 heads appear.

$C = \{(H,T), (T,H), (T,T)\} \leftrightarrow$ At least one tail.

$D = \{(H,H), (H,T), (T,H)\} \leftrightarrow$ At least one head.

$E = \{(H,H), (H,T)\} \leftrightarrow$ First head.

$F = \{(T,H)\} \leftrightarrow$ Only one head, second position.

- Union: "At least one H or at least one T"
  $= C \cup D = S$
- Intersection: "At least one H and at least one T"
  $C \cap D = \{(H,T); (T,H)\}$.
- Mutually Exclusive: "Exactly 2 T and exactly 2 H."
  $A \cap B = \emptyset$.

The definition of probability

- Background experiment.
- S = sample space; $E \subset S$ is an event.
- The mathematical def. of probability below does not apply to all possible subsets of S, if S is a complicated space. However, it does apply to a very large collection of such subsets, denoted by $\mathcal{A}$, as long as $\mathcal{A}$ satisfies the following conditions.

Condition 1 $\quad S \in \mathcal{A}$

Condition 2 $\quad$ If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$

Condition 3 $\quad$ If $A_1, A_2, \ldots, A_m, \ldots \in \mathcal{A}$, then

$$\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}.$$

- In what follows we will call $\mathcal{A}$ the "collection of events".

- Important note If S = finite, then $\mathcal{A}$ = the collection of all subsets of S.

This is Exercise 12, page 16.

- To define probability, we must assign to each event A in our collection of events a number. That is equivalent to saying that we need to define a set function, that is a function the argument of which is a A. This function will be defined on $\mathcal{A}$ and will have a real number as its value. In order for this function to be called a probability function, or probability measure, or simply probability, it must satisfy 3 axioms.

Definition (Probability)

The set function P defined on the collection of events associated with a sample space S is called a probability if:

Axiom 1 For every event A, $P(A) \geq 0$.

Axiom 2 $P(S) = 1$.

Axiom 3 For every sequence of mutually exclusive (disjoint) events $A_1, A_2, \ldots$ we have

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Remark From now on, whenever we use the term "event" it is implicitely assumed that it belongs to $\mathcal{A}$.

Important properties of probability

1. $P(\emptyset) = 0$

2. For every finite sequence of n disjoint events $A_1, \ldots, A_n$ we have

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i).$$

3. $P(A^c) = 1 - P(A)$, for any event A.

4. If $A \subset B$, then $P(A) \leq P(B)$.

5. For any event A, $\boxed{0 \leq P(A) \leq 1.}$

6. For any two events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

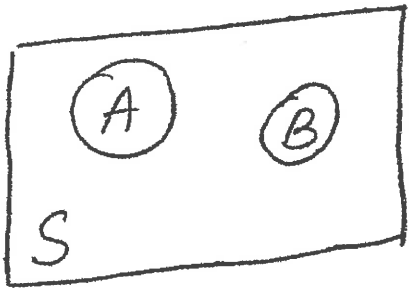
Note See Th. 1.5.6 and 1.5.8 for further properties.

- In 1933, Andrey Kolmogorov published the book "Foundations of the theory of probability", laying the axiomatic foundations of probability as a field.

Example Consider two events A and B s.t. $P(A)=\frac{1}{3}$ and $P(B)=\frac{1}{2}$.

Find $P(B \cap A^c) \overset{\text{Def}}{=\!=} P(B \setminus A)$. (Set difference)

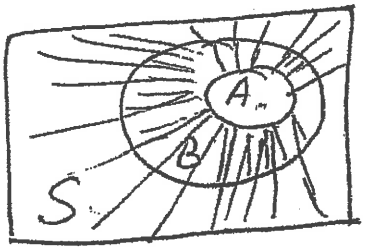
(a) A and B are disjoint. //



If $A \cap B = \emptyset \Rightarrow B \subset A^c \Rightarrow B \cap A^c = B$

$\Rightarrow P(B \cap A^c) = P(B) = \frac{1}{2}$.

(b) $A \subset B$ //



· We can write

$$B = A \cup (B \cap A^c), \text{ where } A \cap (B \cap A^c) = \emptyset$$

Then $P(B) = P(A) + P(B \cap A^c) \Longrightarrow$

$$P(B \cap A^c) = P(B) - P(A) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Thus, when $A \subset B$: $P(B \setminus A) = P(B) - P(A)$.

(c) $P(A \cap B) = \frac{1}{8}$

Recall the "partioning a set" property from page 7 notes, or Th. 1.4.11 book. Then:

$$B = [A \cap B] \cup [A^c \cap B].$$

Therefore: $P(B) = P(A \cap B) + P(A^c \cap B)$, since the sets in the union are disjoint.

This gives:

$$P(A^c \cap B) = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}.$$

Finite sample spaces

- Experiments with only finitely many outcomes
- $S = \{s_1, \ldots, s_n\}$.
- To specify a probability measure on S is to assign to each outcome $s_i \in S$ a probability p_i, $1 \le i \le n$. These numbers must satisfy:

  (1) $p_i \ge 0$, for $i = 1, 2, \ldots, n$.

  (2) $\sum_{i=1}^{n} p_i = 1$.

- If $A \subset S$, $P(A)$ can be found by adding all the probabilities p_i of all the outcomes that correspond to A.

Example (1.6.2)

- We have 5 fibers. They have different lengths, say of 1, 2, 3, 4 and 5 inches, respectively.
- We want to know which fiber breaks first.
- Let S_i = "The fiber with length i breaks first".

$$S = \{S_1, S_2, S_3, S_4, S_5\}.$$

- Assume $p_i =: P(S_i) = \alpha \cdot i$, where α is a proportionality factor to be determined, $\alpha > 0$.

Since $\sum_{i=1}^{5} p_i = 15 \cdot \alpha = 1 \Rightarrow$

$$\alpha = \frac{1}{15}.$$

Therefore
$$\begin{pmatrix} S_1 & S_2 & S_3 & S_4 & S_5 \\ \frac{1}{15} & \frac{2}{15} & \frac{3}{15} & \frac{4}{15} & \frac{5}{15} \end{pmatrix}.$$

- Let A be the event that the fiber that breaks first is no more than 2 inches.

Then: $A = \{ s_1, s_2 \}$ and

$$P(A) = \frac{1}{15} + \frac{2}{15} = \frac{3}{15} = \frac{1}{5}.$$

Finite sample spaces with equally likely outcomes (Simple spaces).

If, in addition to (1) and (2) above, we also have $p_i = p$, for $i = 1, \ldots, n$,

then: $n \cdot p = 1 \Rightarrow p = \frac{1}{n}$.

Therefore, the probability of each $s_1, s_2, \ldots, s_n$ is the same and equals $\frac{1}{n}$.

Rule If A is an event in the simple space and if it contains exactly m outcomes, then:

$$P(A) = \frac{m}{n}, \text{ for } 0 \leq m \leq n.$$

Example 1: Toss two fair coins once:

$$S = \{\underbrace{(T,T)}_{s_1}, \underbrace{(H,H)}_{s_2}, \underbrace{(T,H)}_{s_3}, \underbrace{(H,T)}_{s_4}\}.$$

Fair coins $\Rightarrow$ $P(s_i) = \frac{1}{4}$ for each $i = 1, \ldots, 4$.

$P(\text{"We see exactly one H"}) = P(\{s_3, s_4\})$

$$= \frac{2}{4} = \frac{1}{2}.$$

Reading assignment: Example 1.6.4 in book.

Example 2 Let S be as in Example 1.

Let $E = \{s_2, s_4\}$ = "The first coin lands heads."

$F = \{s_2, s_3\}$ = "The second coin lands heads".

Let G = Either the first or the second is heads.

$= E \cup F$.

Solution: $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$$= \frac{2}{4} + \frac{2}{4} - \frac{1}{4} = \frac{3}{4}.$$

Solution 2 $G = \{S_2, S_3, S_4\} \Rightarrow$

$$P(G) = \frac{3}{4}.$$

NOTE For both solutions, we counted the number of outcomes in the events of interest. Sometimes (many times!) these events can be too large for that, and we need to have a principled way of counting.